

# Alejandro Lopez

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| Open to remote opportunities

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## Professional Summary

Full-stack software engineer with 4+ years building AI-integrated, cloud-native commerce systems (AWS Lambda, EventBridge, Redis, Bedrock) and concurrently architecting a cross-platform Flutter workspace OS as founder and sole developer. Delivered production-grade integrations across 6 marketplace platforms achieving Meta Commerce Partner certification and shipped enterprise security systems with dual-layer E2EE encryption and WebRTC-based real-time communication.

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## Core Competencies

**Languages & Frameworks:** C# | .NET Core | Python | Dart / Flutter | AngularJS | Node.js

**AI & Machine Learning:** AWS Bedrock (Claude, Nova, Titan) | OpenAI API | OpenSearch Serverless (kNN / RRF)

Vector Embeddings | LLM Inference & Prompt Engineering | ETL Pipelines | Apache Spark

**Cloud & DevOps:** AWS Lambda | AWS ECS / Fargate | AWS EventBridge | AWS SQS / SNS  
AWS Secrets Manager | AWS S3 | Firebase | Docker | Redis

**Security & Integrations:** OAuth 2.0 | HMAC-SHA256 | AES-256 / RSA / E2EE | WebRTC (Agora)  
REST API Design | Shopify GraphQL Bulk Operations

**Databases & Tools:** SQL Server (MSSQL) | MongoDB | Firebase Firestore / Realtime Database |  
Redis | GraphQL

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## Professional Experience

### Founder & Software Engineer, Aloz

**May 2022 – Present**

Building a cross-platform application OS in Flutter — a widget-based workspace platform with integrated team communication, AI-powered data tools, and enterprise-grade encryption.

- Architecting a widget-based cross-platform application framework in Flutter, deploying a single codebase across 6 platforms (web, Windows, macOS, Android, iOS, Linux) with drag-and-drop workspace management and Firebase backend.
- Implemented a dual-layer encryption system for team communication — AES-256 field-level encryption and RSA-based end-to-end encryption with platform-specific secure key storage across all supported platforms.
- Engineered a real-time team communication module integrating Agora WebRTC for voice and video, Firebase Realtime Database for live presence, and granular role-based access control with audit logging.
- Integrated OpenAI API for AI-powered data visualization recommendations and automated report generation, enabling natural-language-driven analytics within a credentials-secured, role-scoped workspace environment.

### Software Engineer, GoDataFeed

**Jan 2022 – Present**

Multi-tenant e-commerce SaaS platform enabling merchants to distribute product catalogs across 8+ channels (Amazon, Walmart, Meta, TikTok, Shopify) and synchronize orders at scale.

- Architected an AI product-content generation pipeline on AWS Lambda, EventBridge Scheduler, and Redis (distributed state machine), automating attribute generation across 300K+ SKU merchant catalogs.

- Owned the end-to-end Meta Business Extension (FBE) integration across 9 engineering phases — OAuth token centralization, OMS partner registration, and Uninstall API orchestration — achieving Commerce Partner production certification.
  - Designed a multi-platform order routing engine synchronizing 6 marketplace sources (Amazon, Walmart, eBay, Meta, TikTok) to 4 destination connectors, extending the public API to support item-level partial-shipment tracking.
  - Migrated large-catalog XML imports from buffered MemoryStream to streaming XmlReader, eliminating System.OutOfMemoryException on 1.66GB / 364K-SKU feeds and sustaining daily imports across 12 e-commerce platforms.
  - Built TikTok Shop's full product lifecycle integration (HMAC-SHA256 signing, dual-version API factory, media upload pipeline) and a bidirectional Shopify feed using GraphQL Bulk Operations with delta sync.
  - Shipped 4 versioned releases of GoDataFeed's public Magento 2 extension (Composer/Packagist, GitHub-hosted), adding an extension-free REST API import mode enabling catalog synchronization without plugin installation.
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## Education

### **B.S. Computer Science, Minor in Business & Mathematics**

Nova Southeastern University | May 2022 | GPA: 3.9

*Relevant Coursework:* Data Structures & Algorithms | Cloud Computing | AI & Machine Learning | Database Management

### **Accomplishments:**

- Dean's List (multiple semesters) for academic excellence.
- 1st Place, Healthcare Hackathon (2020): Led AI-driven diabetic care app with predictive analytics for patient monitoring.